SCIENTIST GUIDE

RatioAvis 1.1.1

From the *linear model* to the optimality certificate

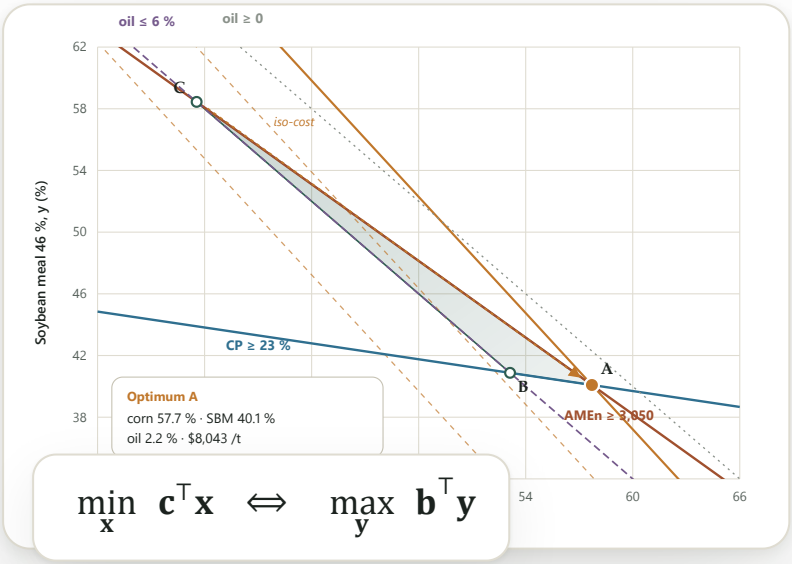
Least-cost broiler diet formulation: method, duality, economic reading, validation against published diets and reproducibility.

1st edition · October 2026 · covers RatioAvis 1.1.1

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Companion to the Producer Guide.



METHOD

Two-phase simplex

GUARANTEE

Strong duality

primal–dual gap of 3.9 × 10^{-16} in the case

VALIDATION

6 / 6

published formulas recovered exactly

REPRODUCIBILITY

Model as JSON

and a SciPy verification script

BEFORE READING

Scope, notation and citation

Who this guide is for, what it assumes you know and how equations are written.

This guide is for those who need to understand, verify or publish what RatioAvis does: researchers, animal-nutrition lecturers, graduate students and journal reviewers. It explains the full mathematical model, how it is solved, why the result is provably optimal and how the program was validated.

It assumes a working knowledge of poultry nutrition and linear algebra. Step-by-step use of the interface is in the Producer Guide; here we use the **Scientist** view and the same worked case: Ross 308, as-hatched, starter (0–10 d), 2,200 m a.s.l., with the −50 kcal/kg AMEn adjustment.

Notation

SYMBOL	MEANING	UNIT
$i \in I$	selected ingredient (decision variable)	—
$j \in J$	nutrient with a requirement (20 possible)	—
x_i	inclusion of ingredient i in the diet	% (kg/100 kg)
c_i	delivered ingredient price	$/kg
a_{ij}	content of nutrient j in ingredient i, as fed	%, kcal/kg, mg/kg
b_j^-, b_j^+	minimum and maximum requirement of j in the diet	unit of j
l_i, u_i	ingredient inclusion limits	%
Z	cost of 100 kg of diet (objective)	$/100 kg
y_k	dual variable (shadow price) of constraint k	$/100 kg per unit
d_i	reduced cost of ingredient i	$/100 kg per % point

Vectors are in bold ($\mathbf{x}$, $\mathbf{c}$) and equations are numbered on the right. Button and tab names appear as on screen: ⚙ Formulate, Formulation › Mathematical model and reproducibility.

Box types

≡ **DEFINITION**

A concept with its exact meaning.

∴ **RESULT**

A theorem or property the program relies on, with its practical consequence.

◆ **IN THE CASE**

The real numbers of the worked case, exactly as the app computes them.

! **CAUTION**

An assumption or limitation that can change a conclusion.

How to cite

“ **SOFTWARE**

Mojica-Zárate, H. T., Barrera-Guzmán, L. Á., & Velázquez-Vázquez, G. (2026). *RatioAvis: Guided least-cost diet formulation for broiler chickens* (Version 1.1.1) [Software]. Zenodo. https://doi.org/10.5281/zenodo.23123050

“ **THIS GUIDE**

Mojica-Zárate, H. T., Barrera-Guzmán, L. Á., & Velázquez-Vázquez, G. (2026). *RatioAvis: Scientist Guide* (1st ed.). Distributed with RatioAvis 1.1.1. https://hectortecumshe-ai.github.io/RatioAvis/instructivo-cientifico.html

hectortecumshe-ai.github.io/RatioAvis

Scan the code to open the app on your phone or share it.

Section R.3 provides a ready-made paragraph for a paper’s *Materials and methods*.

FOUNDATIONS

M The model and its solution

What problem RatioAvis solves, how it solves it and why it can guarantee that no cheaper diet exists.

M.1 The diet problem

Least-cost diet formulation is one of the founding problems of operations research. Stigler (1945) posed the search for the cheapest human diet meeting requirements; Dantzig solved such problems with the simplex method (1947; Dantzig, 1963), and since the 1950s the feed industry has formulated this way every day. RatioAvis implements that classical framework transparently: every number entering and leaving the model is visible.

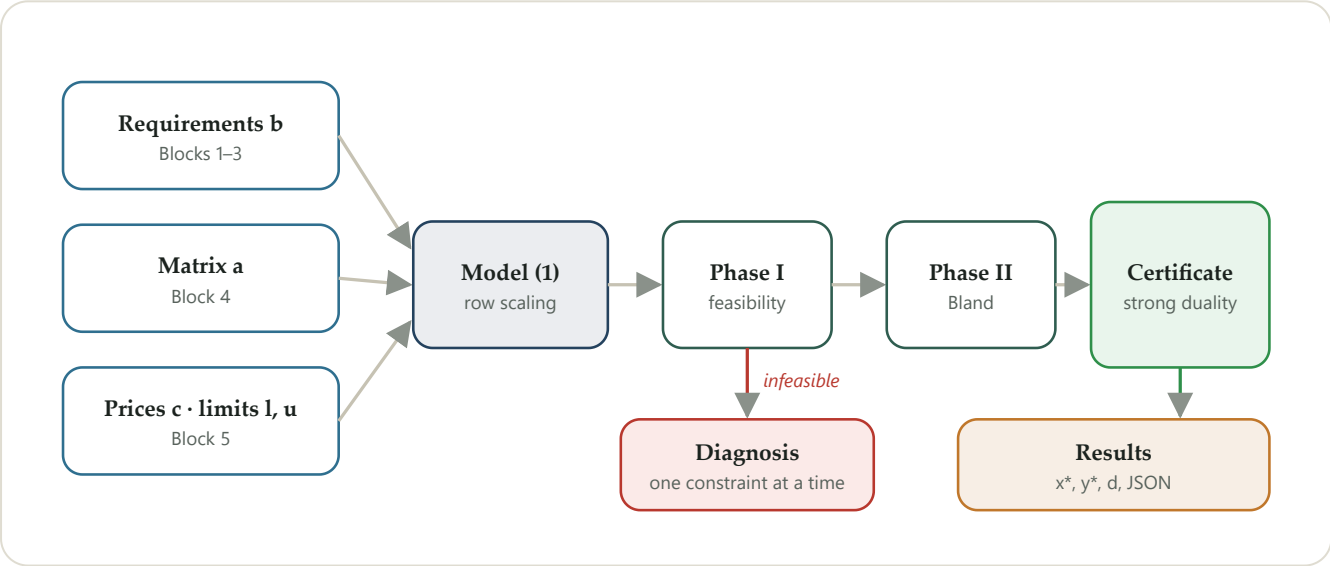


FIGURE 1 **Solver architecture.** Data from blocks 1–5 build the model; phase I seeks feasibility, phase II the optimum, and the duality certificate validates the solution before it is shown. If there is no solution, a diagnosis identifies the conflicting constraint.

M.2 Mathematical formulation

With the selected ingredients I and the nutrients with a requirement J, RatioAvis solves the linear program:

$$
\begin{aligned}
\min_{\mathbf{x}} \quad & Z = \textstyle\sum_{i \in I} c_i x_i \\
\text{subject to} \quad & \textstyle\sum_i x_i = 100 \\
& \textstyle\sum_i a_{ij} x_i \geq 100\, b_j^- \quad \forall j : b_j^- \text{ defined} \\
& \textstyle\sum_i a_{ij} x_i \leq 100\, b_j^+ \quad \forall j : b_j^+ \text{ defined} \\
& l_i \leq x_i \leq u_i, \quad x_i \geq 0
\end{aligned}
\tag{1}
$$

The first constraint closes the blend at 100 %; the next ones impose each nutrient's minimum and, when set, maximum; the last one the inclusion limits. Since x_i is in kg per 100 kg, Z is the cost of 100 kg and the cost per tonne is $10Z$.

≡ **HOW A_{IJ} IS BUILT**

- **Amino acids:** if the genetic line uses a digestible basis, the ingredient's standardized digestible column is used (`dLys`, `dMet`…); on a total basis (NRC 1994), the total one.
- **Electrolyte balance:** computed per ingredient and entered as one more linear nutrient (equation 6).
- **Limits:** a minimum only creates a row if it is above 0, and a maximum only if it is below 100 %.

M.3 The geometry of the optimum

Each constraint is a half-space; their intersection, the **feasible region**, is a convex polyhedron. Iso-cost lines are parallel and the optimum, if it exists, always lies at a **vertex**. With 14 ingredients the polyhedron lives in 14 dimensions, but a reduced three-ingredient model (corn, soybean meal and oil, with oil completing 100 %) can be drawn on the plane:

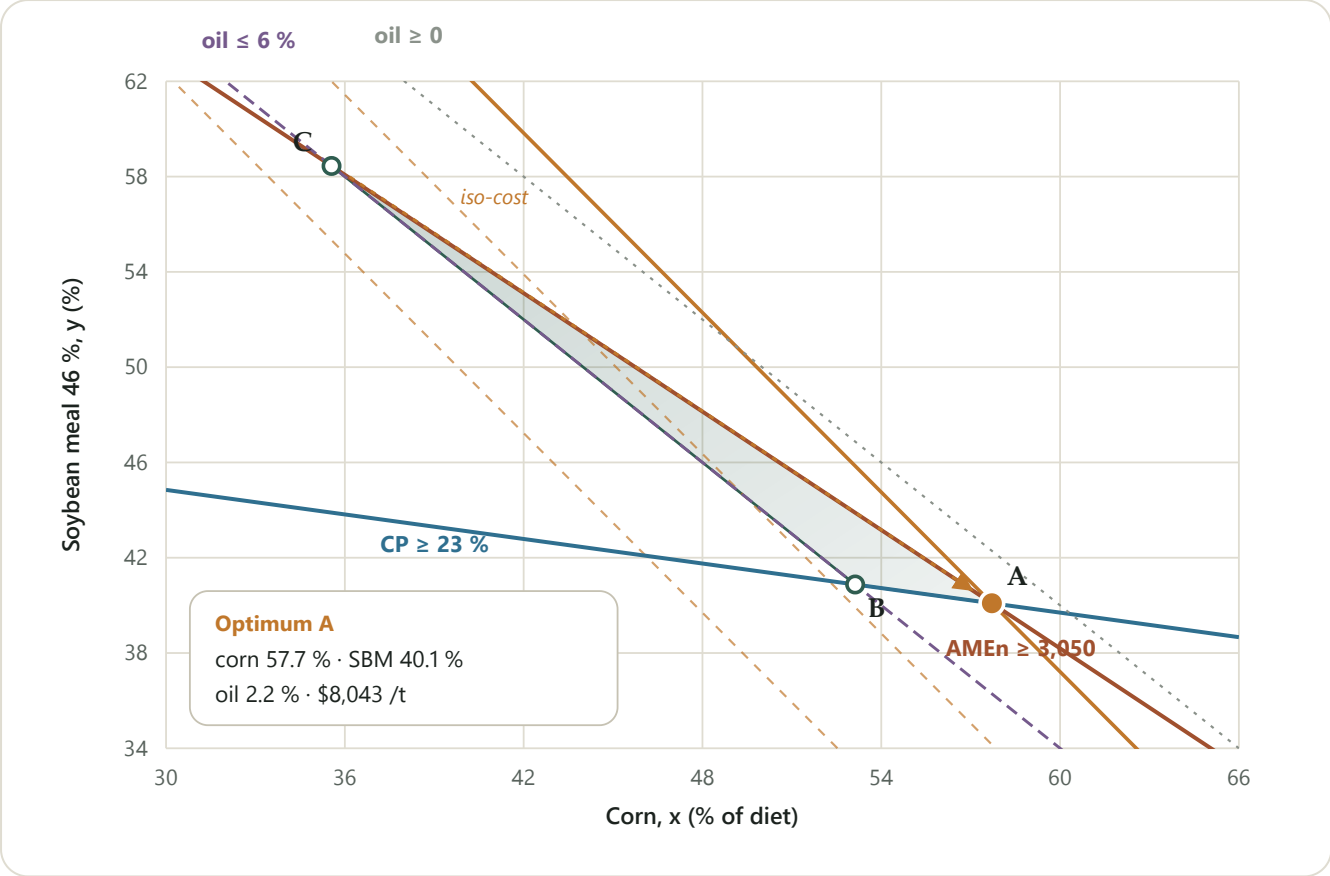


FIGURE 2 **Feasible region of a reduced teaching model.** Corn (x) and soybean meal 46 % (y); oil = 100 − x − y, at most 6 %. Requirements: protein ≥ 23 % and AMEn ≥ 3,050 kcal/kg, using the app's matrix and prices. Simplex moves from vertex to vertex until **A**, where energy and protein cross: 57.7 % corn, 40.1 % soybean meal and 2.2 % oil. It is an illustration; the real case has 14 variables and 39 constraints.

M.4 The two-phase simplex method

RatioAvis brings model (1) to standard form with slack and artificial variables and solves it with its own tableau simplex, written in JavaScript and run in the browser:

1. **Scaling.** Each row is divided by its largest absolute coefficient, so that energy (thousands of kcal) and amino acids (hundredths) share comparable scales. Duals are unscaled at the end.
2. **Phase I.** The sum of artificial variables is minimized. If it is not zero at the end (threshold 10^{-7}), the problem is **infeasible** and the diagnosis runs (M.7).
3. **Phase II.** From that feasible basis the real cost is minimized. The first variable with a negative reduced cost enters and the minimum-ratio row leaves, ties broken by lowest index: **Bland's rule** (1977), which guarantees no cycling even under degeneracy, common in diets because of fixed limits (premix, choline).

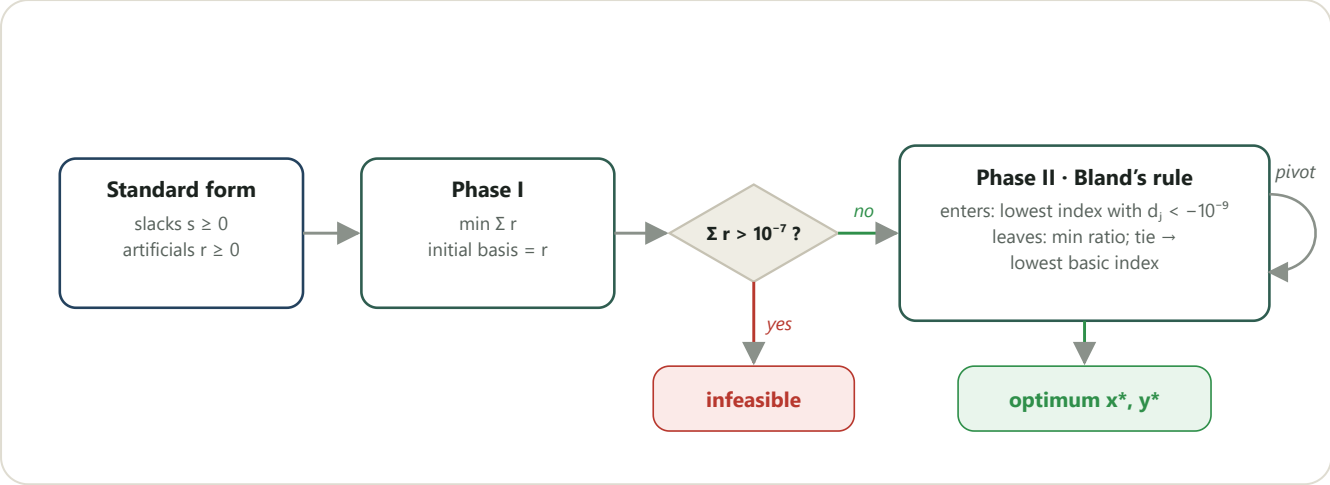


FIGURE 3 **Two-phase simplex flow.** Tolerances are those in the RatioAvis 1.1.1 source code.

M.5 Duality and the optimality certificate

Every linear program has a corresponding **dual problem**. Writing the constraints of (1) as $\mathbf{Ax} \gtrless \mathbf{b}$, the dual seeks the prices $\mathbf{y}$ that maximize the value of the requirements without any ingredient being "undervalued":

$$\max_{\mathbf{y}} \; W = \mathbf{b}^{\top}\mathbf{y} \quad \text{s.t.} \;\; \mathbf{A}^{\top}\mathbf{y} \le \mathbf{c}, \quad y_k \ge 0 \text{ for } \ge \text{ rows}, \; y_k \le 0 \text{ for } \le \text{ rows} \tag{2}$$

∴ **STRONG DUALITY THEOREM**

If $\mathbf{x}^{*}$ is primal feasible, $\mathbf{y}^{*}$ is dual feasible and $\mathbf{c}^{\top}\mathbf{x}^{*} = \mathbf{b}^{\top}\mathbf{y}^{*}$, both are optimal. No feasible diet can cost less than $\mathbf{b}^{\top}\mathbf{y}^{*}$, since every dual feasible solution is a lower bound on cost (weak duality).

RatioAvis does not simply trust the simplex: after every formulation it **recomputes** the three conditions from the final numbers and only marks the solution as certified if the tolerances are met:

CHECK	WHAT IT MEASURES	TOLERANCE	IN THE CASE
Primal feasibility	largest violation of a constraint or of $x \ge 0$	$< 10^{-6}$	4.3×10^{-14}
Dual feasibility	sign of each y and reduced costs ≥ 0	$< 10^{-6}$	4.0×10^{-13}
Primal–dual gap	$\lvert Z - W \rvert \;/\; \max\left(1, \lvert Z \rvert\right)$	$< 10^{-7}$	3.9×10^{-16}

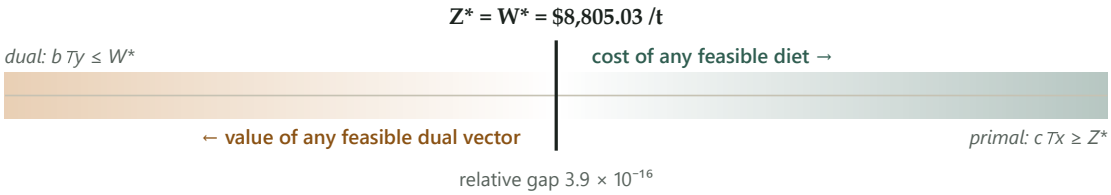


FIGURE 4 **What the certificate guarantees.** Any feasible diet costs at least Z^*; any feasible dual price vector is worth at most W^*. When both meet, the optimum is pinned down: in the case, \$8,805.03 per tonne, with a relative difference of 3.9×10^{-16}, the limit of double-precision arithmetic.

M.6 Shadow prices, reduced costs and entry price

SHADOW PRICE

The dual value y_k of a constraint is the derivative of the optimal cost with respect to its right-hand side, $y_k = \partial Z^* / \partial b_k$, valid while the optimal basis does not change. Only **binding** constraints (met with equality) have a non-zero shadow price: this is **complementary slackness**.

Since the right-hand side of a nutrient in (1) is $100 b_j$ and Z is in \$/100 kg, the app reports in the **Shadow prices** table the change in cost **per tonne**:

$$\frac{\partial \left(10 Z^*\right)}{\partial b_j} = 1000\, y_j \qquad \frac{\partial \left(10 Z^*\right)}{\partial u_i} = 10\, y_i \tag{3}$$

left, per unit of requirement (1 kcal/kg or 1 % point); right, per % point of an inclusion limit.

REDUCED COST AND ENTRY PRICE

The reduced cost $d_i = c_i - \mathbf{a}_i^\top \mathbf{y}$ of an ingredient outside the formula is how much Z would rise per % point it were forced in. If its price fell by d_i, it would sit at the threshold of entering; hence the app reports the **entry price**:

$$c_i^{\text{entry}} = c_i - d_i \quad \left(\$/\text{kg}\right) \tag{4}$$

M.7 Infeasibility diagnosis

When phase I ends with positive artificials, the app runs two tests: (a) it looks for nutrients with a positive minimum that no selected ingredient supplies; (b) it re-solves the model **removing one constraint at a time** and lists those that, on their own, restore feasibility. It is a first-order diagnosis, fast and easy to interpret.

CAUTION

If no single constraint explains the infeasibility, the conflict involves several at once. The app says so, but does not compute an irreducible infeasible subsystem (IIS): requirements must be relaxed or ingredients added in a reasoned way.

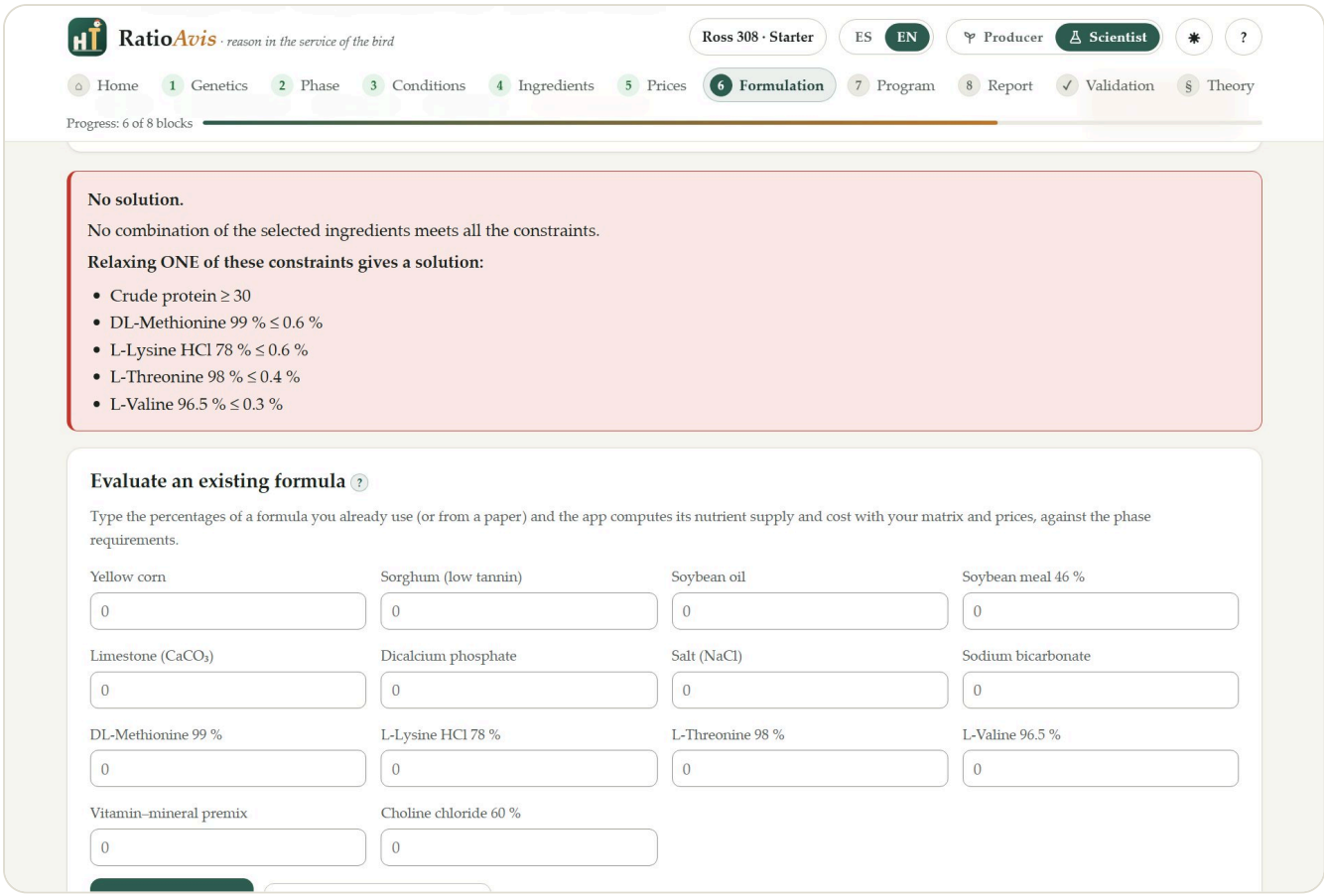


FIGURE 5 **Diagnosis in the scientist view.** With protein ≥ 30 % (impossible with these ingredients), five constraints are each enough on their own to restore feasibility.

BLOCKS 1 TO 5 · INPUTS

1–5 The model's data

Where b, a, c, l and u come from, which adjustments the app applies and which assumptions to declare.

1.1 Requirements b: sources and adjustments

PROFILE	SOURCE	AA BASIS	STATUS
Ross 308 · Arbor Acres Plus	Aviagen (2022)	digestible	current official
Cobb 500	Cobb-Vantress (2022)	digestible	check
Hubbard Efficiency Plus	Hubbard (2022)	digestible	current official
Slow-growing	ME and CP from Usturoi et al. (2023); AA by ideal protein	digestible	guideline
NRC	NRC (1994)	total	historical
Custom	NASEM (2025), Rostagno et al. (2024) or other	digestible	user

On top of the phase table, the app applies explicit, documented adjustments:

$$
\begin{aligned}
&b_{\overline{\mathrm{AA}}} \leftarrow f_{\mathrm{AA}}\, b_{\overline{\mathrm{AA}}}, \quad b_{\overline{\mathrm{PC}}} \leftarrow f_{\mathrm{PC}}\, b_{\overline{\mathrm{PC}}} \qquad (f_{\mathrm{AA}}, f_{\mathrm{PC}}) = \{\,(1.03, 1.02), (0.96, 0.98)\,\}\\
&b_{\mathrm{EM}} \leftarrow b_{\mathrm{EM}} - 50 \qquad \text{if altitude} \geq 1{,}500 \text{ m (when the recommendation is applied)}\\
&b_{\mathrm{DEB}} = [230, 300]\,\mathrm{mEq/kg} \qquad \text{if temperature} \geq 30\ ^{\circ}\mathrm{C}\\
&b_{\overline{\mathrm{Xan}}} = \text{xanthophyll target (70 mg/kg by default) in the last two phases for pigmented}
\end{aligned}
\tag{5}
$$

The first factor pair is for males and the second for females; as-hatched birds use the table as is.

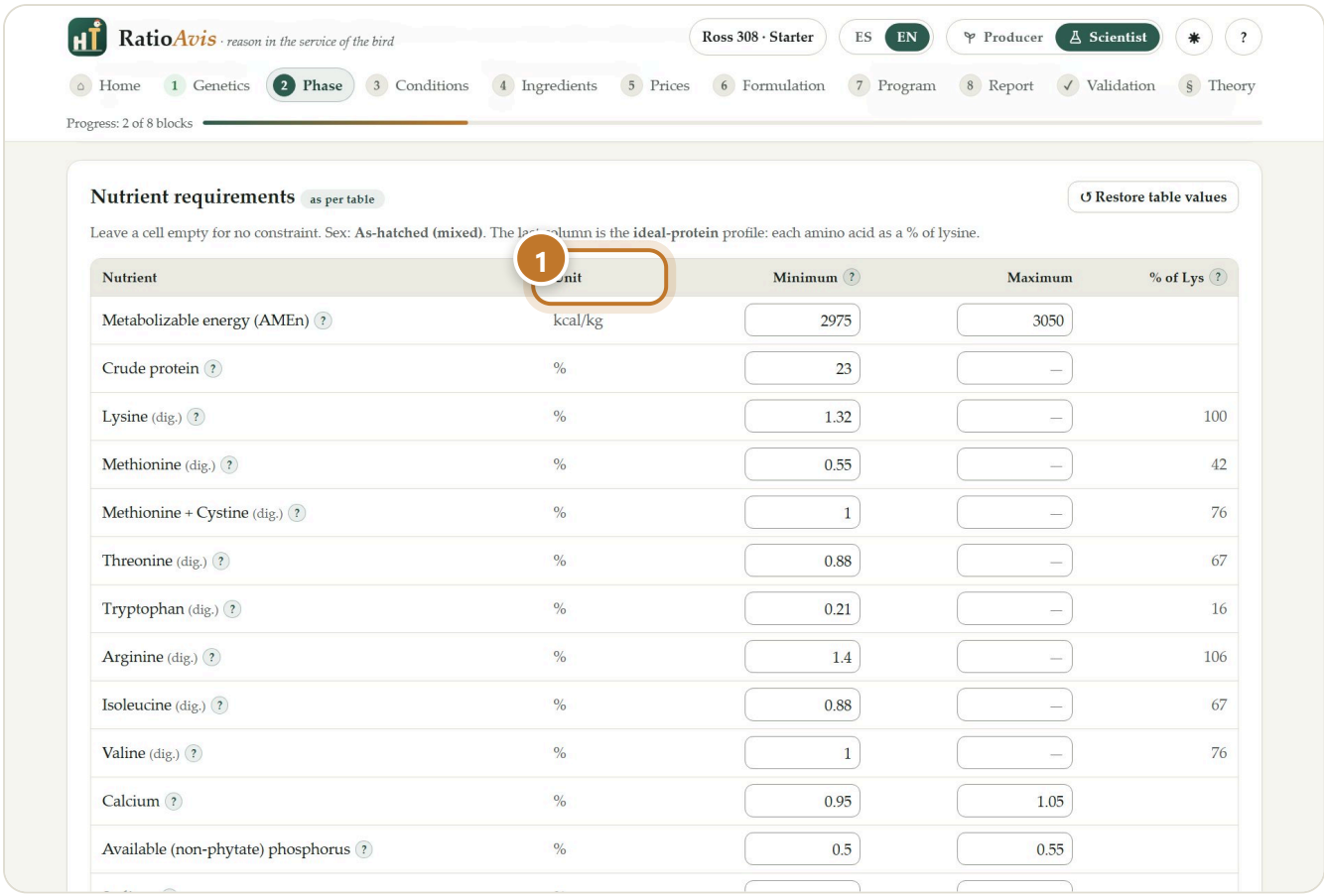


FIGURE 6 **Requirements in the scientist view.** All 20 nutrients and the ideal-protein column: each amino acid as % of lysine.

1 **% of Lys**: the ideal-protein profile of the chosen table.

4.1 The matrix *a*

The library holds **29 ingredients** in eight categories, each with 28 columns: dry matter, AMEn, protein, ether extract, fibre, Ca, available P, Na, Cl, K, linoleic acid, eight total amino acids, their eight standardized digestible values and xanthophylls. Everything is on an as-fed basis (≈ 88–90 % dry matter). Sources are Rostagno et al. (2017), FEDNA (2019) and NRC (1994).

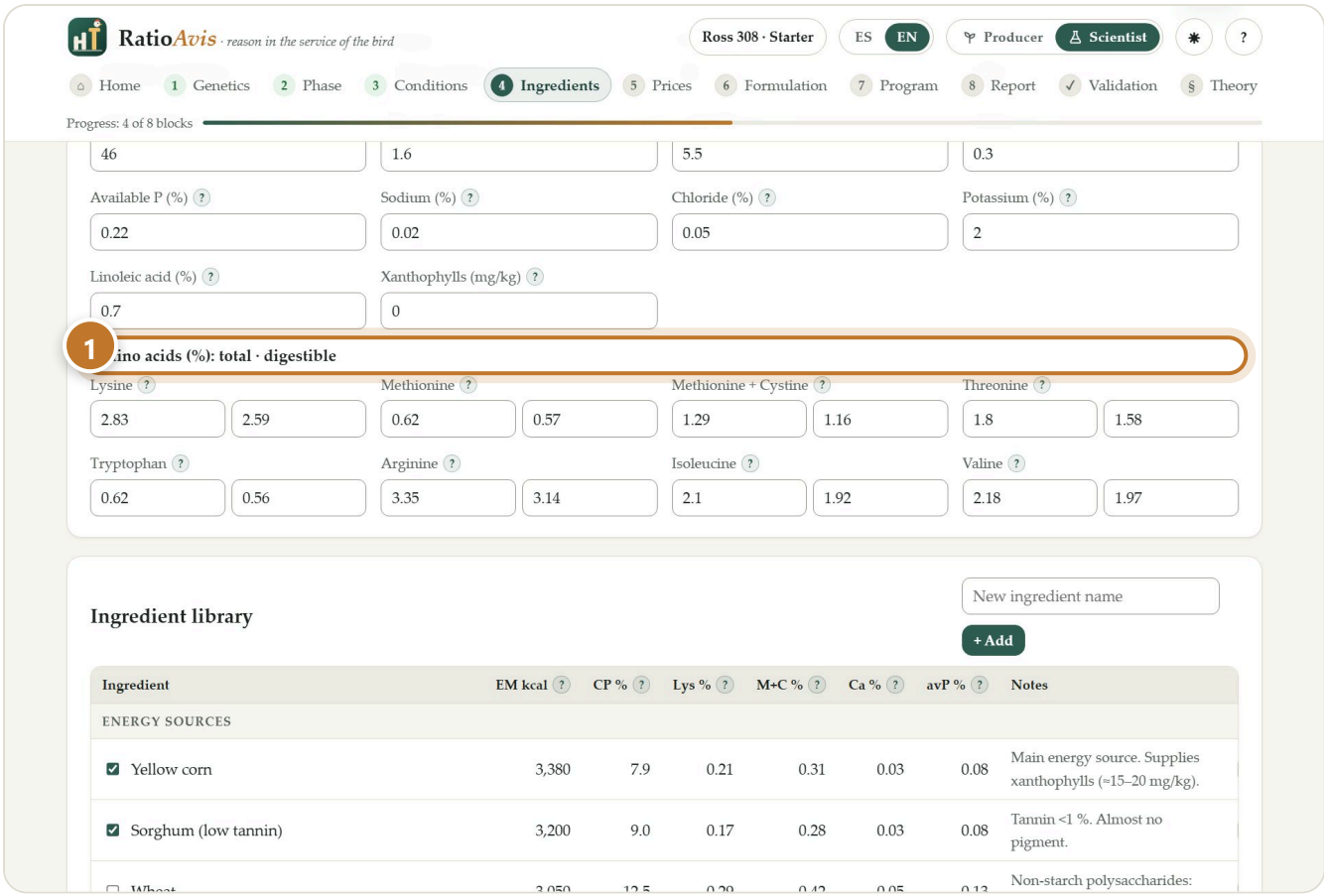


FIGURE 7 **Editable composition of soybean meal 46 %.** The scientist view shows and edits each ingredient's total and digestible amino acids.

1 Each amino acid has two fields: **total · digestible**.

Each ingredient's electrolyte balance (Mongin, 1981) is computed from the atomic masses of Na, K and Cl:

$$\mathrm{DEB} = \frac{\mathrm{Na}\,\% \cdot 10^4}{22.99} + \frac{\mathrm{K}\,\% \cdot 10^4}{39.10} - \frac{\mathrm{Cl}\,\% \cdot 10^4}{35.45} \quad (\mathrm{mEq/kg}) \tag{6}$$

! **CAUTION · THE MATRIX DOMINATES THE ERROR**

The calculation is exact; the uncertainty lies in composition. As the validation shows (chapter V), 5–10 % amino-acid differences between tables are common. For publication, report the matrix used (the JSON includes it) and, if possible, analyse the main ingredients.

5.1 Prices c and limits l, u

The preloaded prices are illustrative. Block 5 also shows two purchasing-efficiency indices:

$$\frac{\$}{1000\text{kcal}} = \frac{c_i}{\text{EM}_i} \cdot 1000 \qquad \frac{\$}{\text{kg PC}} = \frac{c_i}{\text{PC}_i \,/\, 100} \tag{7}$$

They help rank sources but do not replace the model: an ingredient can be the cheapest energy source and still be left out because of its other nutrients. In the case, sorghum supplies energy at \$1.56 versus \$1.63 for corn and does not enter (section 6.4).

BLOCK 6 · FORMULATION

6 A scientific reading of a solution

The worked case solved and interpreted: which constraints drive it, what each is worth and how robust the formula is.

6.1 The solution and its certificate

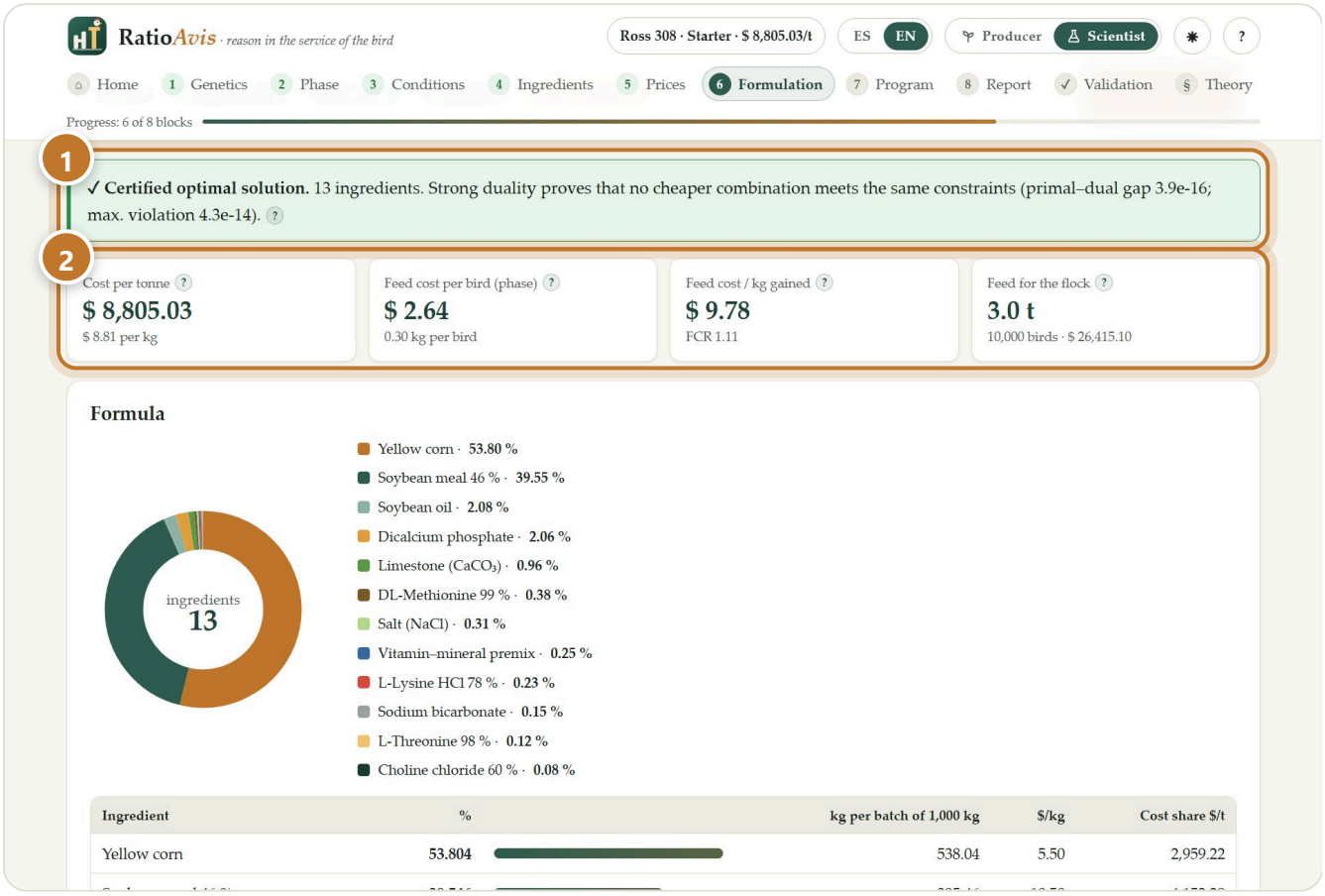
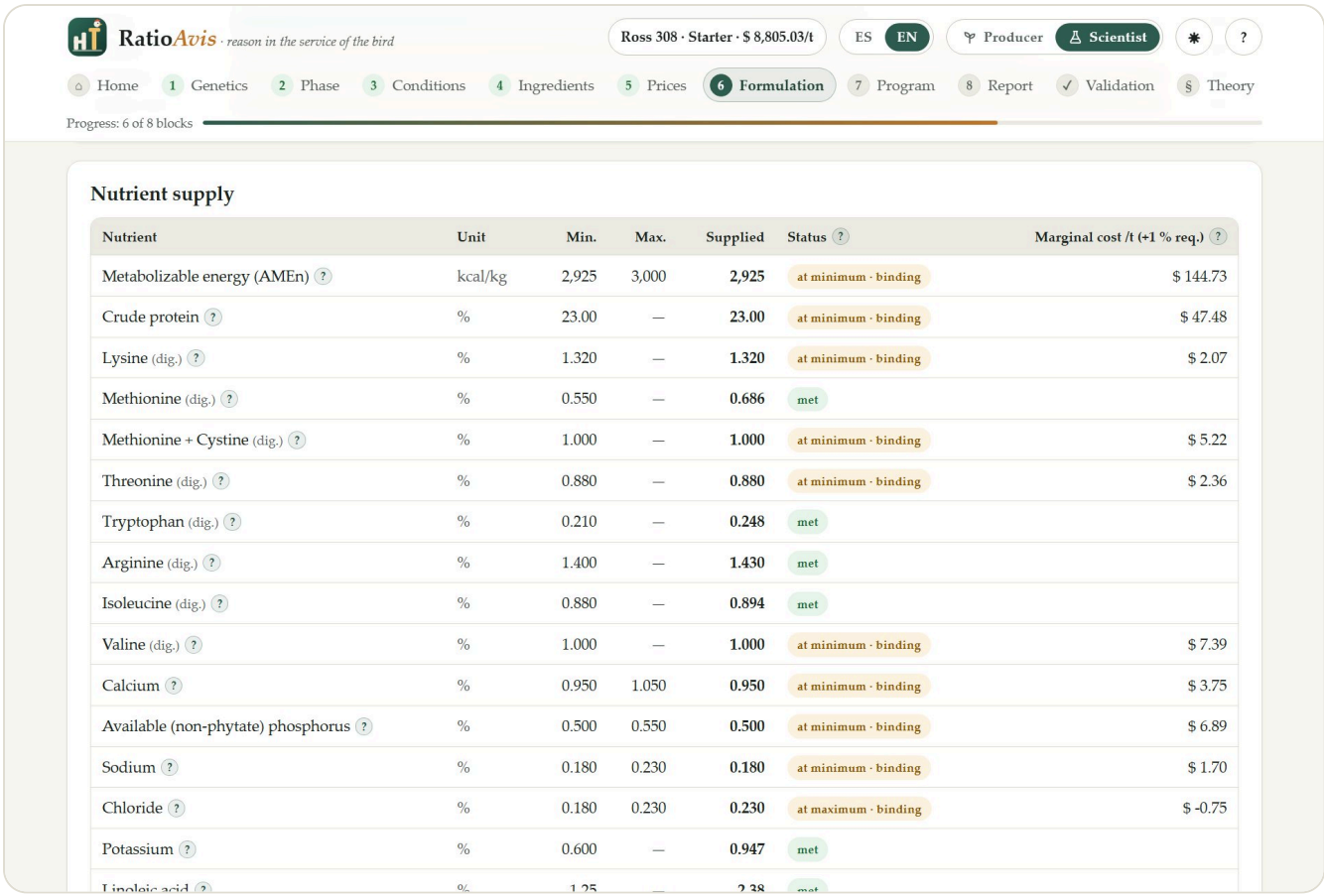


FIGURE 8 **Result in the scientist view.** Instead of the producer message, the certificate appears, with the primal–dual gap and the largest violation.

1. **Certified optimal solution** by strong duality.
2. Economic indicators of the phase.

VARIABLES	CONSTRAINTS	PIVOTS	OPTIMAL COST
14	39	51	$ 8,805.03
ingredients	15 binding	phases I and II	per tonne



Nutrient supply

Nutrient	Unit	Min.	Max.	Supplied	Status	Marginal cost /t (+1 % req.)
Metabolizable energy (AMEn)	kcal/kg	2,925	3,000	**2,925**	at minimum · binding	$ 144.73
Crude protein	%	23.00	—	**23.00**	at minimum · binding	$ 47.48
Lysine (dig.)	%	1.320	—	**1.320**	at minimum · binding	$ 2.07
Methionine (dig.)	%	0.550	—	**0.686**	met	
Methionine + Cystine (dig.)	%	1.000	—	**1.000**	at minimum · binding	$ 5.22
Threonine (dig.)	%	0.880	—	**0.880**	at minimum · binding	$ 2.36
Tryptophan (dig.)	%	0.210	—	**0.248**	met	
Arginine (dig.)	%	1.400	—	**1.430**	met	
Isoleucine (dig.)	%	0.880	—	**0.894**	met	
Valine (dig.)	%	1.000	—	**1.000**	at minimum · binding	$ 7.39
Calcium	%	0.950	1.050	**0.950**	at minimum · binding	$ 3.75
Available (non-phytate) phosphorus	%	0.500	0.550	**0.500**	at minimum · binding	$ 6.89
Sodium	%	0.180	0.230	**0.180**	at minimum · binding	$ 1.70
Chloride	%	0.180	0.230	**0.230**	at maximum · binding	$ -0.75
Potassium	%	0.600	—	**0.947**	met	
Linoleic acid	%	1.25	—	**2.38**	met	

FIGURE 9 **Nutrient supply.** The *Status* column flags nutrients at their limit ("binding") and the last column gives their marginal cost per tonne if the requirement rose by 1 %.

6.2 Which constraints drive the cost

Fifteen constraints are binding: the 100 % sum, nine nutrient minimums (AMEn, protein, lysine, Met + Cys, threonine, valine, Ca, available P and Na), the chloride maximum and the four limits fixing premix and choline. Shadow prices per unit of requirement are not comparable (kcal versus % points); to compare them, they are expressed as the cost of raising each requirement by **1 % relative**, $1000 y_j \cdot b_j / 100$:

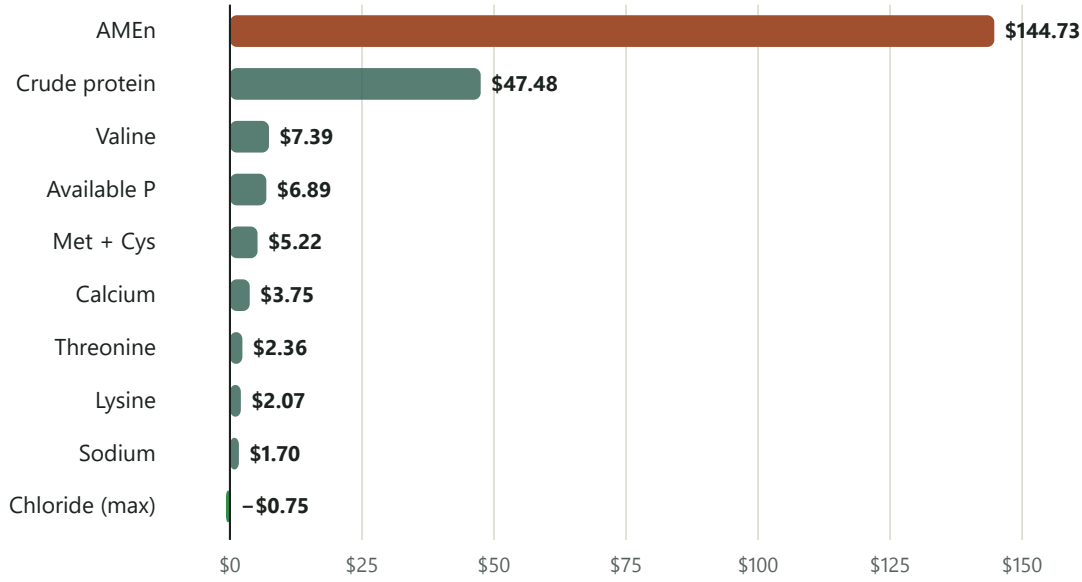


FIGURE 10 **Cost of tightening each binding requirement by 1 %** ($/t). Energy dominates: raising it 1 % (29 kcal) adds $144.7/t, three times more than protein. Chloride is a maximum: relaxing it makes the diet cheaper.

◆ **IN THE CASE · THE SHADOW PRICE PREDICTS THE SAVING**

The energy shadow price is **$4.948 per tonne per kcal/kg**. By (3), lowering the requirement by 50 kcal should save 50 × 4.948 = **$247.40/t**. Formulating with and without the adjustment gives $9,052.43 and $8,805.03: a difference of **$247.40**, exactly as predicted. The optimal basis did not change over that interval, which is why the derivative holds for a 50 kcal change.

Shadow prices in the case (app table) and their reading

BINDING CONSTRAINT	∂Z/∂B · $/T	READING
AMEn ≥ 2,925 kcal/kg	4.948	+1 kcal/kg → +$4.95/t
Crude protein ≥ 23 %	206.432	+0.1 point → +$20.64/t
Dig. lysine ≥ 1.32 %	156.583	+0.01 point → +$1.57/t
Met + Cis ≥ 1.00 %	522.293	+0.01 point → +$5.22/t
Dig. threonine ≥ 0.88 %	267.642	+0.01 point → +$2.68/t
Dig. valine ≥ 1.00 %	738.980	+0.01 point → +$7.39/t
Calcium ≥ 0.95 %	394.540	+0.01 point → +$3.95/t
Available P ≥ 0.50 %	1,378.480	+0.01 point → +$13.78/t
Sodium ≥ 0.18 %	944.167	+0.01 point → +$9.44/t
Chloride ≤ 0.23 %	−326.888	+0.01 point of ceiling → −$3.27/t

The chloride sign explains a choice that may look surprising at first: the app uses **sodium bicarbonate** (0.153 %) besides salt, because sodium is binding and chloride is already at its ceiling. Bicarbonate supplies sodium without chloride.

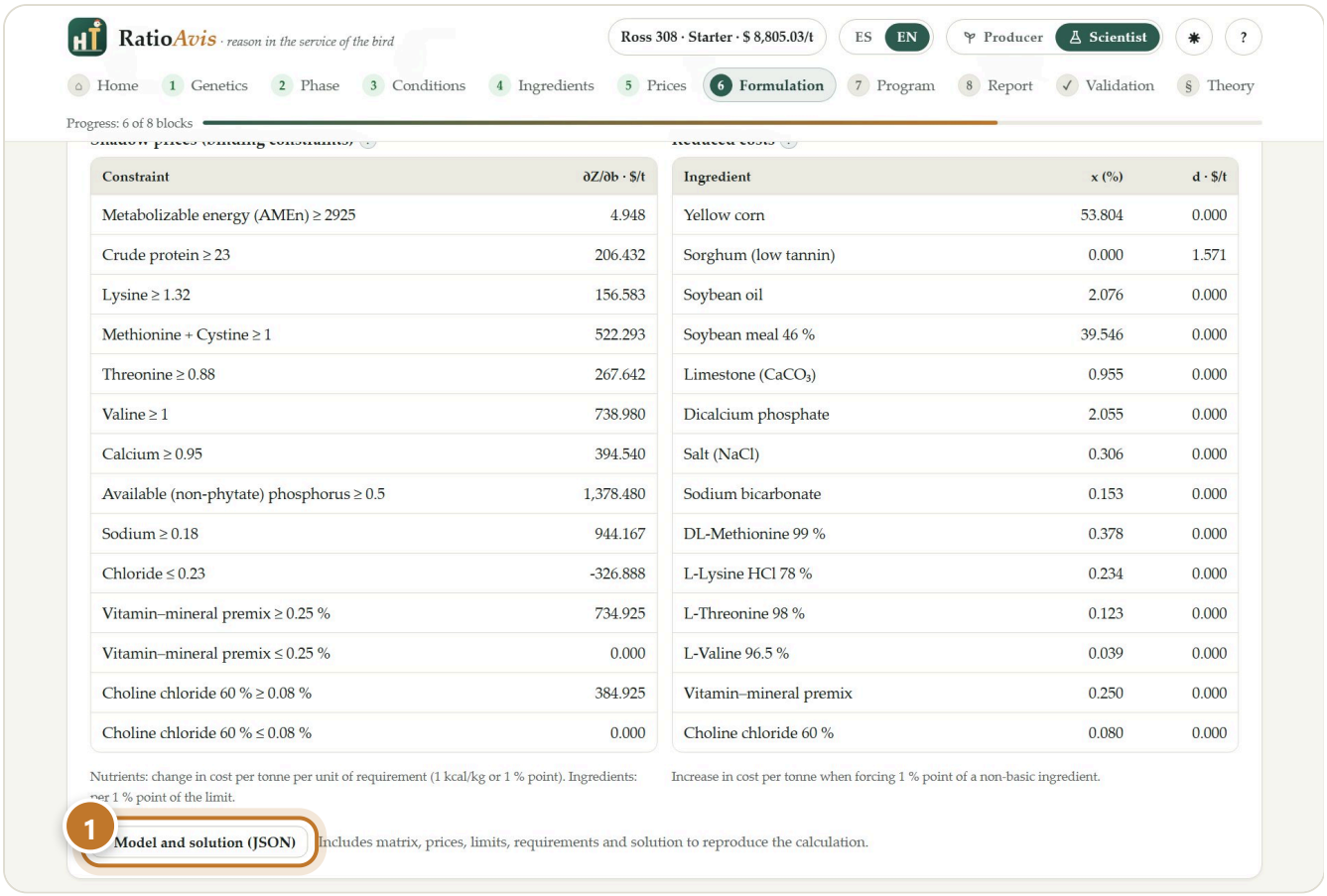


FIGURE 11 **Shadow prices and reduced costs in the app,** with the full-model export.

1 **Model and solution (JSON)**: matrix, prices, limits, requirements and solution (chapter R).

6.3 Where the cost sits

An ingredient's share of the diet and its share of the cost can differ widely. Comparing both guides purchasing better than price per kilo:

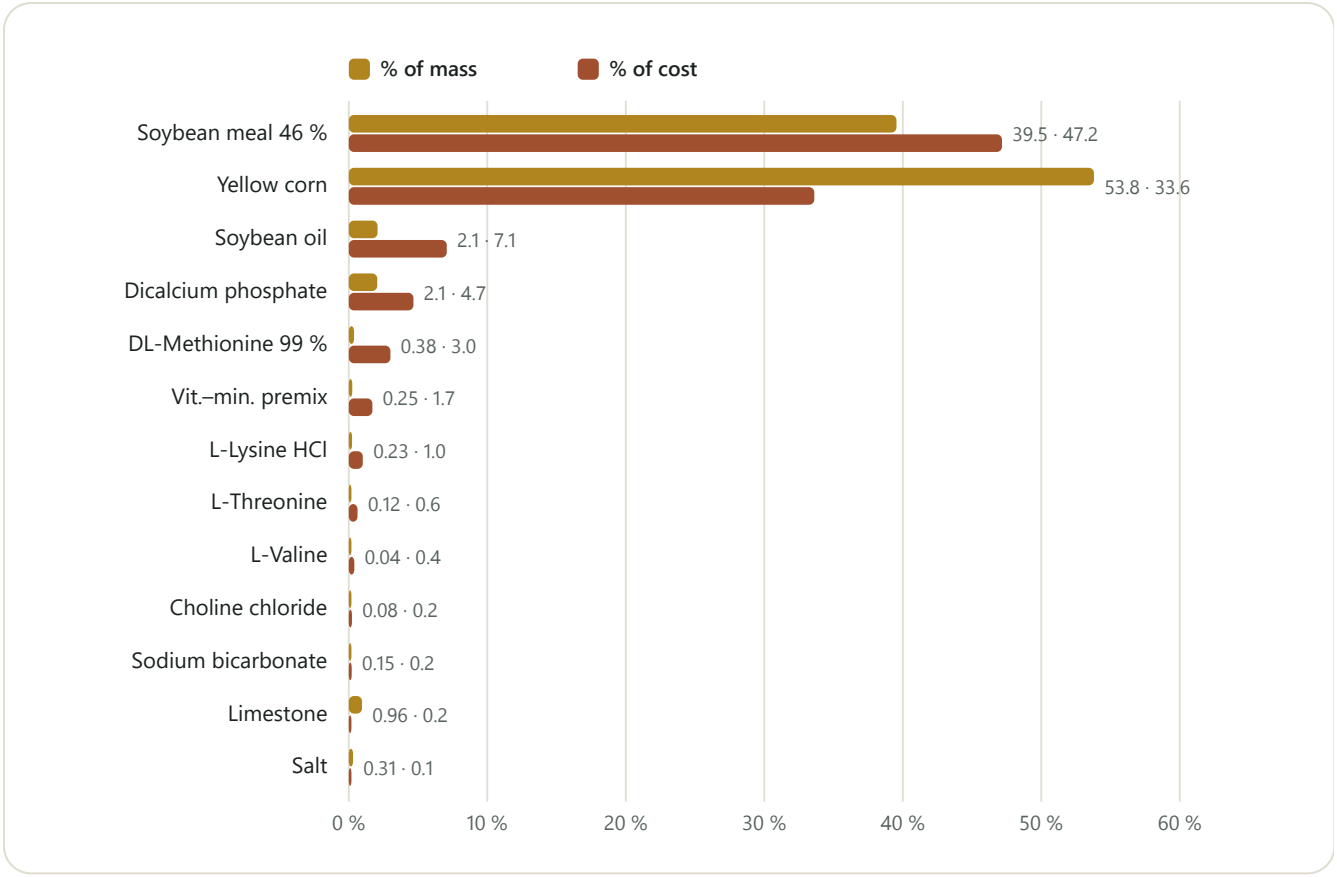


FIGURE 12 **Share of mass versus share of cost** in the case's optimal diet. Soybean meal is 39.5 % of the mass and 47.2 % of the cost; DL-methionine is 0.38 % of the mass and 3.0 % of the cost.

6.4 Reduced costs: the sorghum case

Of the 14 ingredients, only sorghum is left out. Its reduced cost is **1.571 $/t per % point**, i.e., 0.1571 $/kg. By (4), its entry price is 5.00 − 0.157 = **$4.84/kg**: if the supplier offers it at that price or less, reformulating pays. This is the number the Producer view translates as "it would start to pay off".

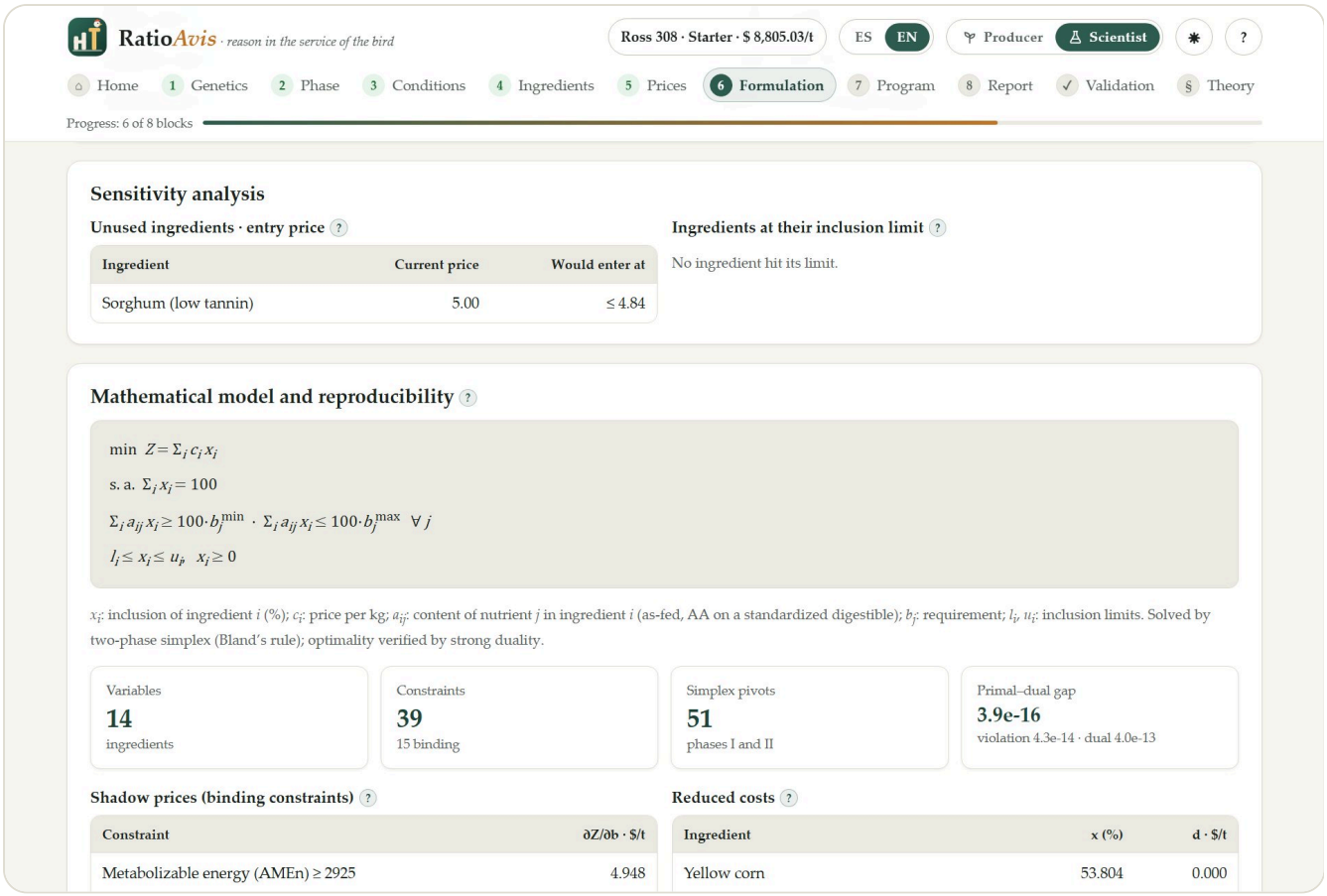


FIGURE 13 **Sensitivity analysis.** Entry price of unused ingredients and effect of relaxing binding inclusion limits.

6.5 Formula quality: ideal protein and indicators

The optimum meets the minimums, but some ratios the model does not impose as constraints are worth checking. The app computes them under **Quality indicators**:

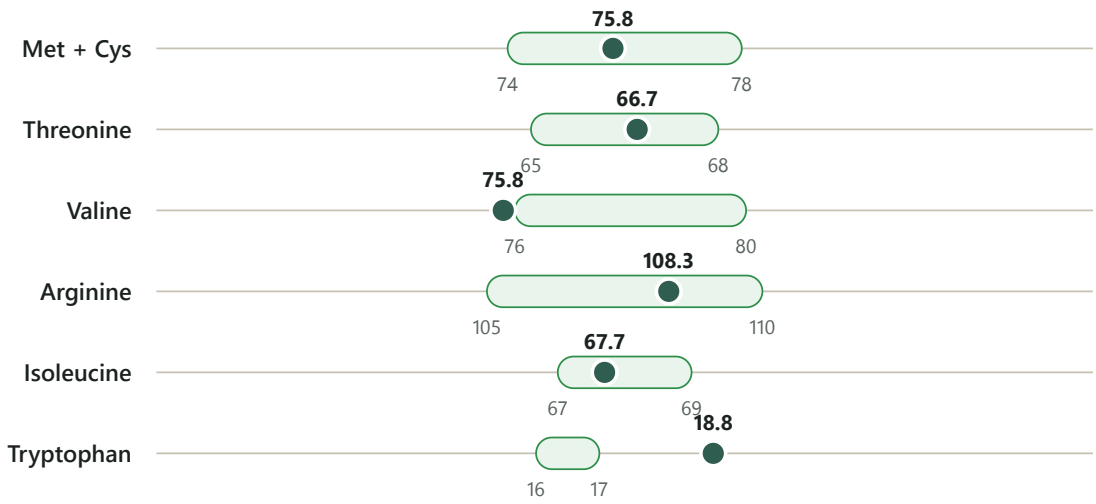


FIGURE 14 **Ideal-protein profile of the case diet** (digestible amino acids as % of lysine) against the reference ranges used by the app. All fall within or above their range; valine (75.8) sits at the lower edge, as it is the amino acid the app had to supplement.

INDICATOR	CASE	REFERENCE
Ca : available P	1.90 : 1	1.8–2.1 : 1
Electrolyte balance	256 mEq/kg	200–300 (optimum ≈ 250)
ME : CP	127 kcal/%	informative
Total fat (EE)	4.63 %	< 7 % for good pellets

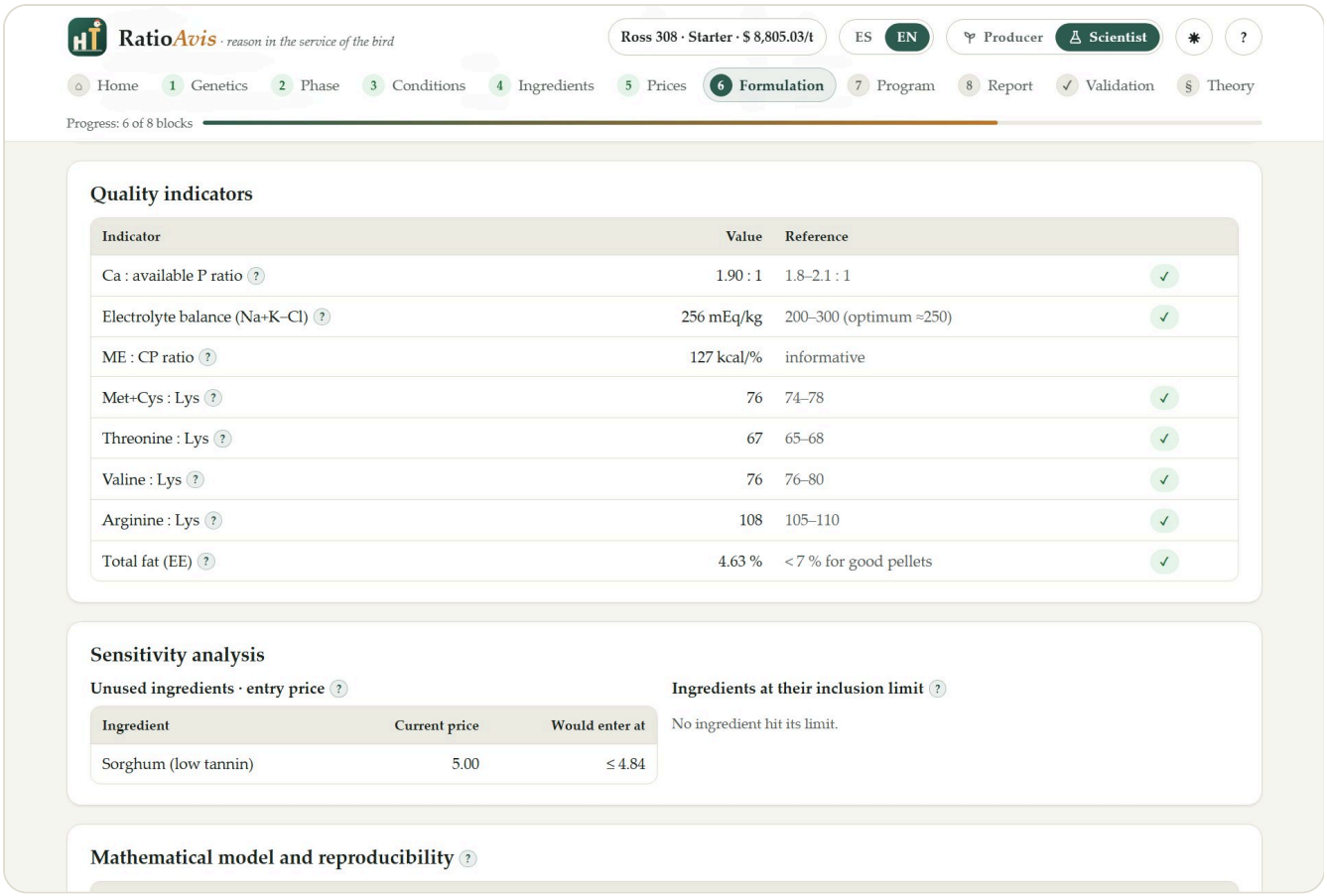


FIGURE 15 **Quality indicators in the app.**

BLOCKS 7 AND 8

7–8 Multiphase program and report

How the full program is built and what the report provides for a log book or a paper.

7.1 Program: one model per phase

Block 7 solves an independent model (1) for each phase of the line, with the same ingredients and prices and the adjustments of (5) for sex, pigmentation and heat. The cycle cost per bird and per kilo of live weight is:

$$C_{\text{ave}} = \sum_{f} q_f \, p_f^{*} \qquad C_{\text{kg PV}} = \frac{C_{\text{ave}}}{\sum_f g_f} \tag{8}$$

with q_f the intake per bird in the phase (kg), p_f^{*} the optimal cost per kg and g_f the weight gain. In the case: $52.94 per bird and $15.04 per kg of live weight.

> **! CAUTION · SEPARABILITY ASSUMPTION**
>
> Each phase is optimized separately with fixed intake and gain: the program does not model the productive response to nutrient density. Manual edits in Block 2 and the altitude adjustment only apply to the single-phase formulation; hence the starter appears here at $9,052/t.

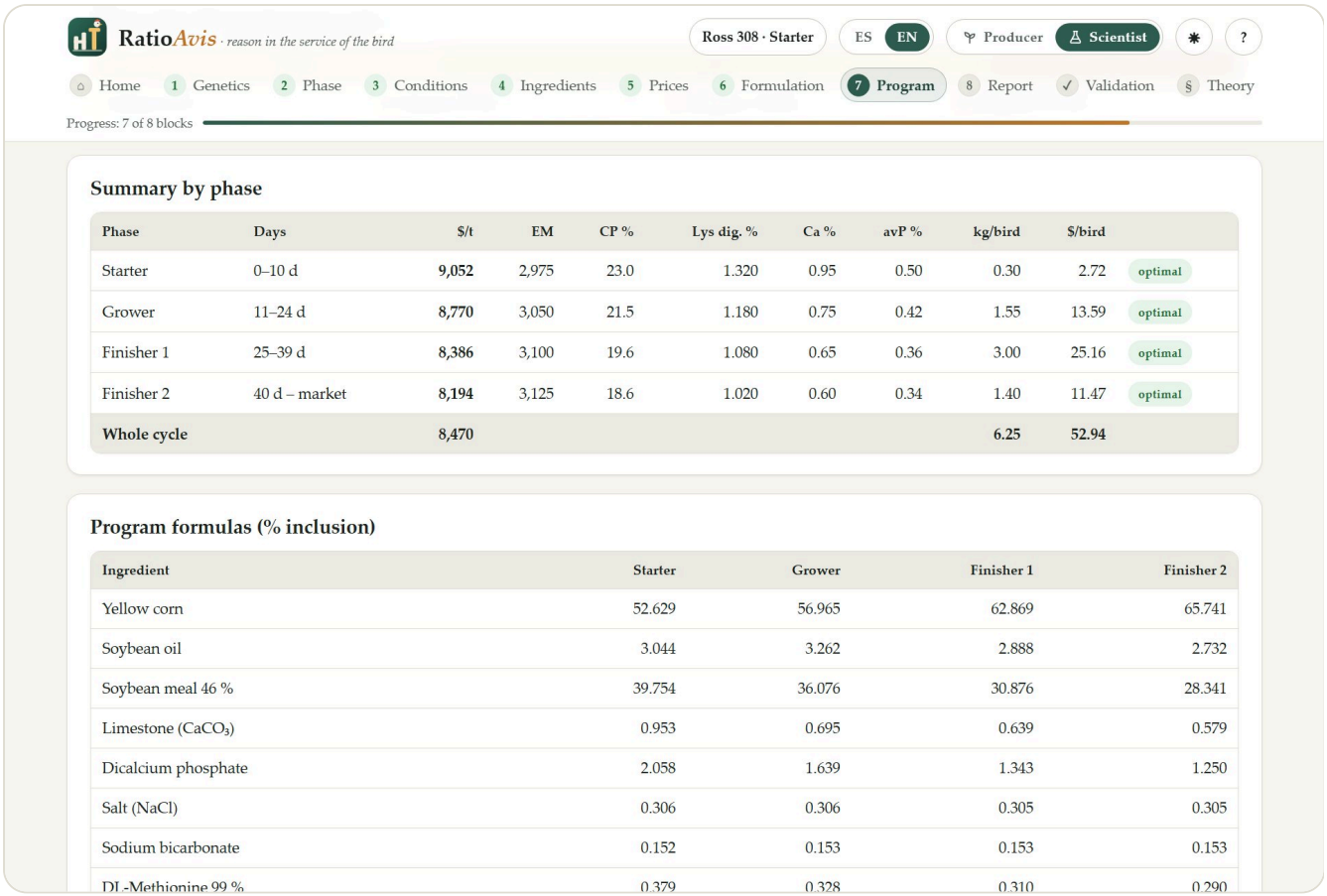


FIGURE 16 **Summary by phase and program formulas.** Every phase with an optimality certificate.

8.1 The scientific report

In the scientist view, the report writes up the method (line, phase, requirement source, amino-acid basis, matrices, cost, supply and binding constraints) and adds the software citation. It is a faithful draft of what was done, useful for a log book or as a starting point for a paper.

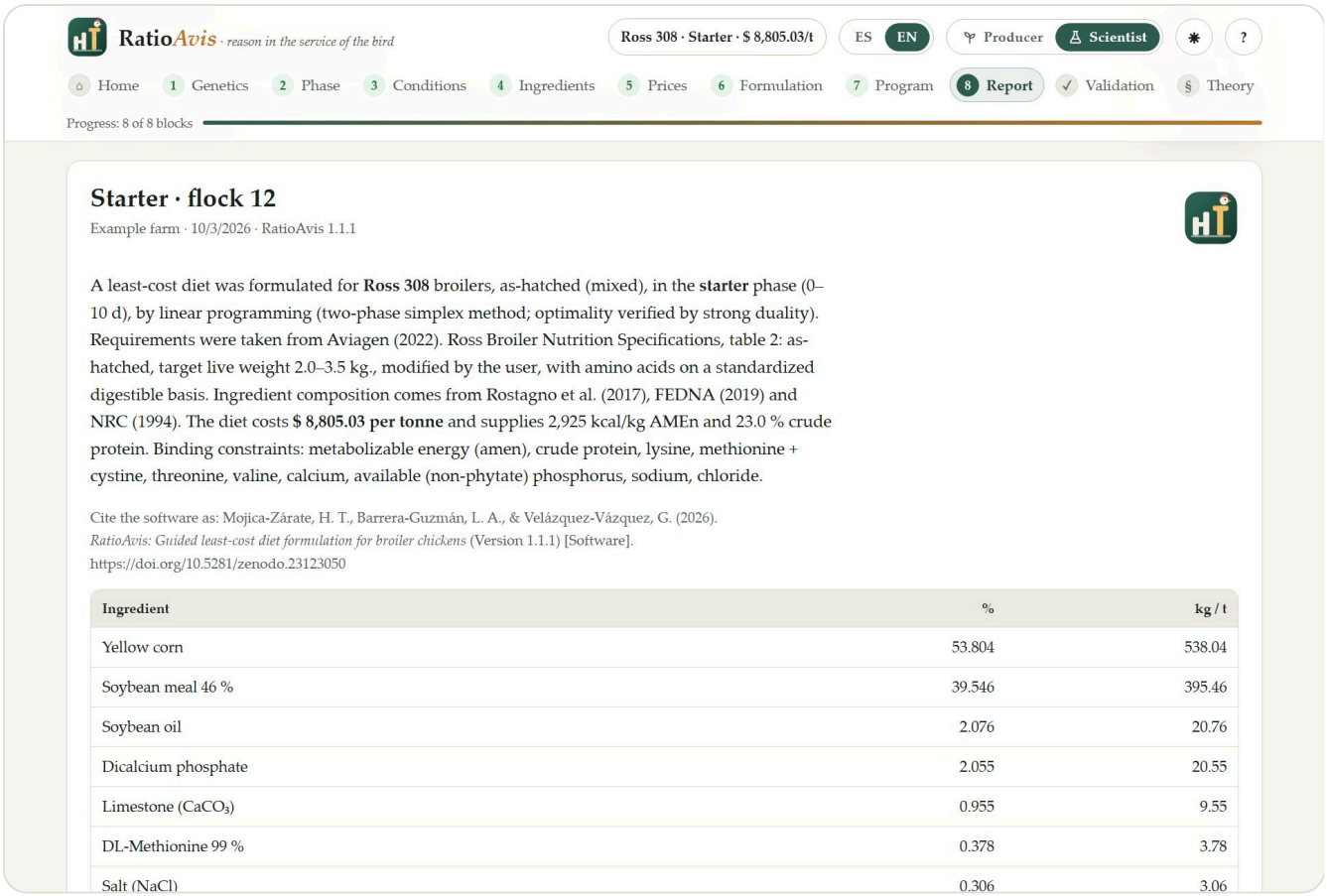


FIGURE 17 **Written report in the scientist view,** with binding constraints and the software reference.

PLATFORM EVALUATION

V Validation

Three independent tests: supply calculation, the optimizer and the solver's mathematics.

V.1 Validation design

Eight diets from 4 peer-reviewed papers were used (Mexico, Jordan and the United States; Ross 308, Cobb 500 and Ross 708; total and digestible basis). Each formula was entered as published, normalized to 100 % when the table did not add up exactly, and analysed with the app's library matrix and prices.

1. **Supply calculation.** The calculated analysis reported by the authors is compared with the app's, using the mean absolute percentage error (MAPE).
2. **Optimizer.** The app is asked for the least-cost diet supplying at least what it computes for the published formula, with the same ingredients and their fixed additives. Since the published formula is already feasible, the optimum must match or beat it.
3. **Mathematics.** The simplex is compared with an independent reference solver on random problems.

$$\mathrm{MAPE} = \frac{100}{n} \sum_{k=1}^{n} \frac{\left| v_k^{\mathrm{app}} - v_k^{\mathrm{art}} \right|}{v_k^{\mathrm{art}}} \tag{9}$$

V.2 Test 1: supply calculation

METABOLIZABLE ENERGY	CRUDE PROTEIN	AMINO ACIDS	CA, AVAIL. P, NA
0.8 %	**2.8 %**	**7.6 %**	**6.7 %**
n = 6 · max 1.7 %	n = 8 · max 7.7 %	n = 31 · max 25.5 %	n = 14 · max 11.5 %

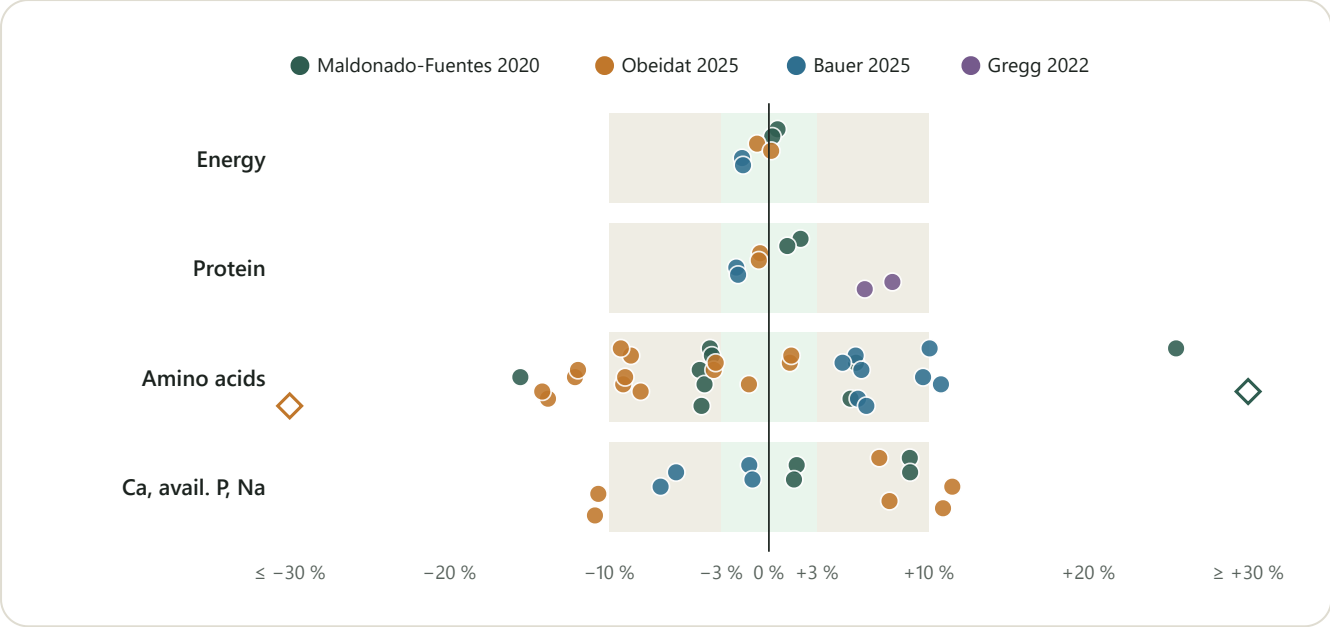


FIGURE 18 **Relative difference between the analysis computed by the app and that reported by the authors,** in 61 comparisons from 8 diets (59 enter the MAPE). The green band marks ±3 %, the usual spread between tables for energy and protein; the grey one ±10 %, the usual spread for amino acids and minerals. Hollow diamonds are probable typos in the paper, excluded from the MAPE.

Energy and protein agree within the normal spread between composition tables. The largest differences occur in amino acids and minerals, and have concrete explanations: soybean-meal protein not reported in two papers (46 % and 47 % were assumed), total versus digestible basis, and different source tables. The app's calculation is an exact weighted sum, so these differences reflect the **matrix**, not the algorithm.

✓ **TYPOS DETECTED**

The comparison also flagged two reported values that look like typos: methionine 0.47 % in the finisher of Maldonado-Fuentes et al. (2020), while the same table reports 0.66 % for diets with the same DL-methionine; and valine 1.39 % in the starter of Obeidat et al. (2025), impossible with that formula, while the next phase reports 0.35 %.

V.3 Test 2: the optimizer

In the 6 diets with enough reported values (at least four comparable nutrients), the app recovered **exactly the published formula**, with differences below 0.005 points for every ingredient and an optimality certificate. In other words, with those ingredients and that supply, each published diet was already the least-cost one.

PAPER · PHASE	PUBLISHED ($/T)	APP ($/T)	SAME
Maldonado-Fuentes et al. (2020) · Starter	8,982.85	**8,982.83**	✓
Maldonado-Fuentes et al. (2020) · Finisher	9,345.90	**9,345.88**	✓
Obeidat et al. (2025) · Starter	8,395.21	**8,395.19**	✓
Obeidat et al. (2025) · Grower	8,573.13	**8,573.11**	✓
Bauer et al. (2025) · Pre-starter	7,916.56	**7,916.54**	✓
Bauer et al. (2025) · Starter	7,777.62	**7,777.60**	✓

Costs at the library's example prices. The app comes out 2 cents lower because the test requires the supply with a relative tolerance of 10^{-6}.

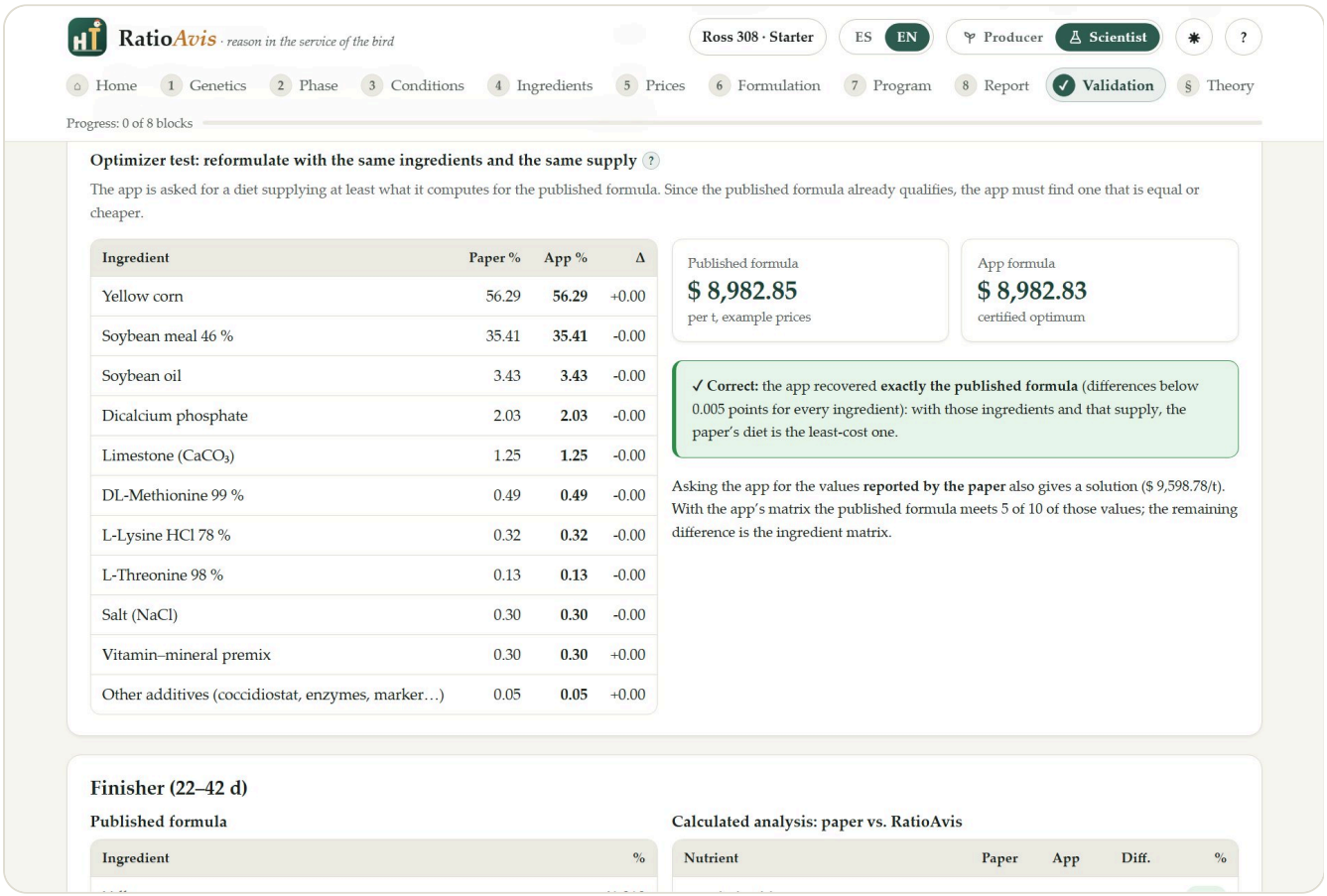


FIGURE 19 **Optimizer test in the app** for the starter of Maldonado-Fuentes et al. (2020): published versus app formula, ingredient by ingredient.

V.4 Test 3: comparison with HiGHS

The app's simplex was compared with HiGHS (Huangfu & Hall, 2018), the linear-programming solver used by SciPy, on **400 randomly generated problems** built from the 6 requirement tables and the ingredient library, varying requirements (±8 %), available ingredients and prices:

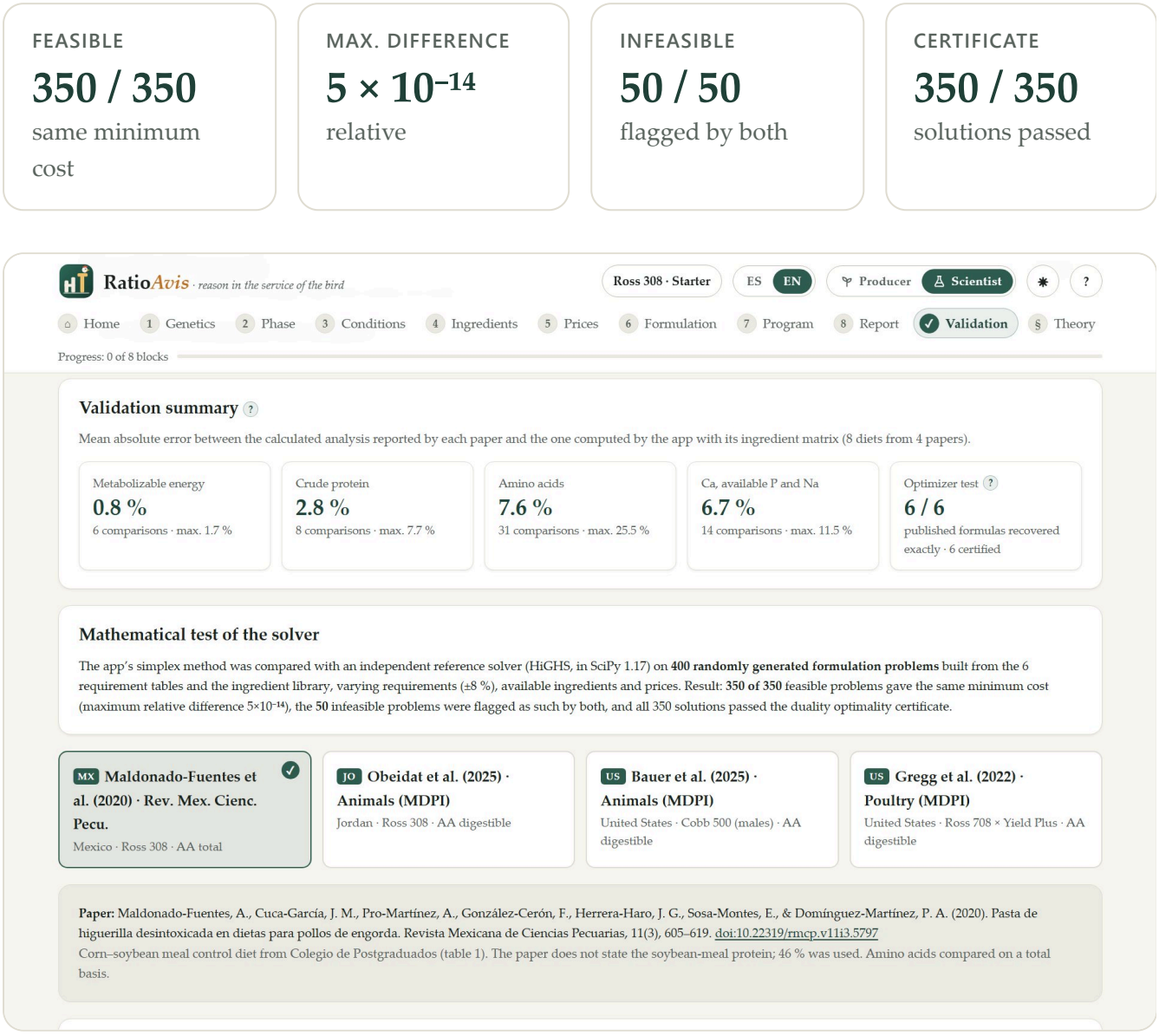


FIGURE 20 **Validation summary in the app,** recomputed every time the block is opened.

OPEN SCIENCE

Reproducibility and publication

How to export the model, verify it independently and describe it in a paper.

R.1 The model file (JSON)

The **⬇ Model and solution (JSON)** button saves everything needed to reproduce the calculation without the app:

FIELD	CONTENT
`software`, `doi`, `fecha`	program version, DOI and calculation date
`genetica`, `sexo`, `fase`, `base`	context; `base` is `dig` or `total`
`requerimientos`	`{nutrient: [min, max]}`, `null` when unbounded
`ingredientes[]`	`id`, `nombre`, `precio`, `min`, `max` and `comp` (the 28 matrix columns)
`solucion`	cost per kg, formula, supply, constraints with duals, certificate and model size

R.2 Independent verification with SciPy

This script reads the JSON, rebuilds model (1) and solves it with HiGHS. With the case model exported from the app it prints **$8,805.03 per tonne** and the same formula, ingredient by ingredient (tested with Python 3.12, NumPy 2.5 and SciPy 1.18).

verificar_ratioavis.py · Python ≥ 3.9 · SciPy ≥ 1.9

```
import json
import numpy as np
from scipy.optimize import linprog

m = json.load(open("dieta_modelo.json", encoding="utf-8"))
ings, req = m["ingredientes"], m["requerimientos"]
dig = m["base"] == "dig"
AA = {"Lys", "Met", "MC", "Thr", "Trp", "Arg", "Ile", "Val"}

def valor(c, k):
    # balance electrolítico (mEq/kg) o columna de la matriz
    if k == "DEB":
        return c["Na"]*1e4/22.99 + c["K"]*1e4/39.10 - c["Cl"]*1e4/35.45
    return c.get("d" + k if (dig and k in AA) else k, 0)

c = np.array([i["precio"] for i in ings])
A_ub, b_ub = [], []
for k, (lo, hi) in req.items():
    a = np.array([valor(i["comp"], k) for i in ings])
    if lo is not None: A_ub.append(-a); b_ub.append(-100 * lo)
    if hi is not None: A_ub.append(a);  b_ub.append(100 * hi)
bounds = [(i["min"] or 0, 100 if i["max"] in (None, "") else i["max"]) for i in ings]

r = linprog(c, A_ub=A_ub, b_ub=b_ub, A_eq=[np.ones(len(ings))], b_eq=[100],
            bounds=bounds, method="highs")
print("Costo por tonelada:", round(r.fun * 10, 2))
for i, x in zip(ings, r.x):
    if x > 1e-7:
        print(f"{i['nombre']:<34}{x:9.3f} %")
```

✓ **GOING FURTHER**

HiGHS returns duals in `r.ineqlin.marginals`. Multiplied by 1000, and with the sign flipped for minimum rows (which the script writes as ≤ with a negative sign), they match the app's **∂Z/∂b · $/t** column. This was verified for the case with SciPy 1.18: all ten nutrient shadow prices match to the third decimal.

R.3 Text for *Materials and methods*

A base paragraph to describe the use of RatioAvis in a paper. Adjust the bracketed data:

Experimental diets were formulated at least cost by linear programming with RatioAvis version 1.1.1 (Mojica-Zárate et al., 2026), which solves the model with the two-phase simplex method under Bland's rule and verifies the optimality of every solution by strong duality (relative primal–dual gap < 10^{-7}). Nutrient requirements followed [Aviagen (2022), Ross 308, starter phase from 0 to 10 d, as-hatched birds], with amino acids on a standardized digestible basis. Ingredient composition was taken from Rostagno et al. (2017) and FEDNA (2019) [or: from laboratory analyses of the batches used], and inclusion limits of [describe] were imposed. The full model and solution were exported in JSON format and are provided as supplementary material.

R.4 Solver technical sheet

METHOD	**Two-phase tableau simplex**	PIVOTING	**Bland's rule (anti-cycling)**
SCALING	**Per row, by max |a|**	ARITHMETIC	**Double precision (IEEE 754)**
PIVOT TOLERANCE	**10^{-9}**	INFEASIBILITY	**Σ artificials > 10^{-7}**
BINDING	**$\|lhs - b\| < 10^{-6} \cdot \max(1, \|b\|)$**	MAX. ITERATIONS	**50,000**
CERTIFICATE	**viol. < 10^{-6} · dual < 10^{-6} · gap < 10^{-7}**	RUNTIME	**Browser, no server**

RESPONSIBLE USE

Assumptions and limitations

What the model assumes and what it does not do, so it can be stated honestly.

- **Linearity and additivity.** The blend's supply is the weighted sum of its ingredients. Interactions are not modelled (for example, the lower ME of fat in chicks is reflected only if the matrix is edited).
- **Deterministic composition.** The matrix is taken as exact; there is no stochastic formulation or automatic safety margin for variability. If needed, add them to the requirement.
- **Ratios not imposed.** Ca:P, ME:CP and ideal-protein ratios are reported as indicators, not constraints. They can be imposed indirectly through minimums and maximums.
- **No response model.** Intake and gain are inputs, not outputs: the app does not predict performance under density changes.
- **Local sensitivity.** Shadow prices and reduced costs are derivatives at the optimum; they hold while the basis does not change. For large changes, reformulate.
- **Decision support.** Results should be reviewed by a qualified animal nutritionist before manufacturing.

REFERENCE

A Appendices

Spanish–English technical glossary and references.

A.1 Spanish–English glossary

Optimal basis *Base óptima*	Set of basic variables at the optimum; while it does not change, shadow prices and reduced costs stay constant.
Primal–dual gap *Brecha primal–dual*	Relative difference between diet cost and dual value; zero at the optimum.
Reduced cost *Costo reducido*	Cost increase per % point a non-basic ingredient is forced in.
Degeneracy *Degeneración*	Vertex with a basic variable at zero; may cause cycling without an anti-cycling rule.
Standardized ileal digestible (SID) *Digestible ileal estandarizado (DIE)*	Fraction of the amino acid absorbed by the end of the ileum, corrected for basal endogenous losses.
Feasibility *Factibilidad*	Existence of at least one diet meeting every constraint.
Complementary slackness *Holgura complementaria*	A non-binding constraint has a zero shadow price, and vice versa.
Entry price *Precio de entrada*	Price below which an unused ingredient would enter the formula: $c - d$.
Shadow price *Precio sombra*	Dual variable: change in optimal cost per unit of a constraint's right-hand side.
Linear programming *Programación lineal*	Optimization of a linear function subject to linear constraints.
Ideal protein *Proteína ideal*	Essential amino-acid profile expressed as % of lysine.
Bland's rule *Regla de Bland*	Lowest-index pivoting rule that prevents cycling in simplex.

Binding constraint *Restricción activa*	Constraint met with equality at the optimum.
Artificial variable *Variable artificial*	Auxiliary phase-I variable used to obtain an initial feasible basis.

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RatioAvis · Scientist Guide, 1st edition, October 2026. Covers RatioAvis 1.1.1. Screenshots come from the real app; charts are original and generated from the app's numerical results for the case described. Prices are illustrative.

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